Mechanisms of Action Prediction

A Multi-Model Ensemble for Highly Imbalanced Multi-Label Classification

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Project Work in Machine Learning and Data Mining

Kaggle LISH MoA Prediction Challenge [1]

The Problem

Prediction task

Given the gene-expression and cell-viability profile of human cells exposed to a compound, predict which of 206 mechanisms of action (MoAs) the compound triggers.

- **Multi-label**: a sample can have no positive label, one, or several.
- **Metric**: mean binary log loss over all samples and targets (lower is better).
- **Difficulty**: very sparse positives, strong target imbalance, and *repeated measurements of the same compound*.

Central methodological question

Does the validation score estimate generalisation to compounds that were not seen in training?

Project objectives

- Reproducible cross-validated pipeline; compare neural and tree models; seed bagging.
- Compound-aware validation, and a measure of how optimistic row-wise CV is.
- Tuning whose epoch selection never uses the scored fold (candidates picked on fold 0, adoption decided on fold 1); ensemble re-validated by nested CV.

Data and Label Structure

Training set

- 23,814 rows: 21,948 treated and 1,866 control (7.8%).
- 772 gene-expression (`g-*`) and 100 cell-viability (`c-*`) features.
- `cp_time` (24/48/72 h), `cp_dose` (D1/D2).
- 3,288 distinct compounds among treated rows (median 6 rows, maximum 718), from `train_drug.csv`.

Labels

- Positive rate: **0.373%** of treated-row/target cells.
- 34.2% of treated rows have no label, 57.1% one, 8.7% two or more.
- Positives per target: 1–832 (median 38); 53 targets have fewer than 20.

- Controls have no MoA by definition: excluded from training, predicted as all zeros at inference.
- The test set has **no compound identifier**: overlap with training can only be inferred indirectly.

Runs and Notation

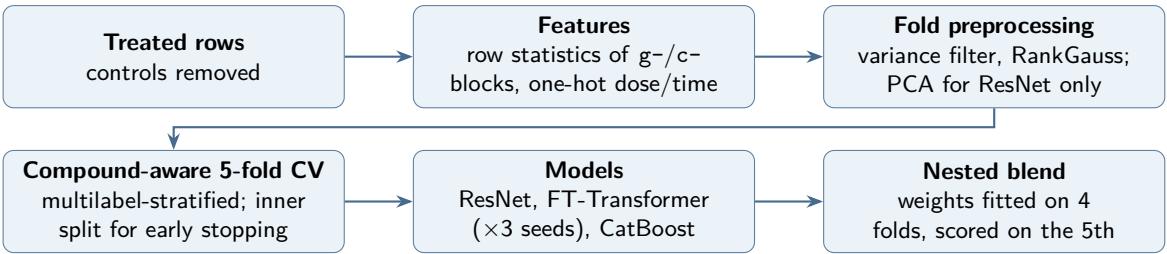
Results refer to the following training runs (all with 5 folds and 3-seed bagging):

Run	CV protocol	Hyperparameters	Kaggle	Time (2×T4)
R	row-wise	defaults, earlier code	yes	≈130 min
R3	row-wise	tuned	no	117 min
H2	hybrid (compound-aware)	tuned	yes	95 min
G3	grouped (compound-aware)	tuned	no	90 min

- **R3, H2, G3: same code and same tuned hyperparameters, only the CV protocol differs**: the controlled comparison (slide 13). R is not part of it.
- H0 and G0: earlier hybrid and grouped runs with default hyperparameters (nested blend 0.01720 and 0.01957), used only in the diagnostics and in the tuned-versus-default comparison.
- **Deliverables** (Python notebooks): training (and two copies for the other protocols), inference, tuning, fold diagnostics, offline blend, test-overlap analysis, nearest-neighbour override test.
- Training and inference are separate: the training notebook saves the artifacts, the inference notebook writes `submission.csv` without retraining.

R used an earlier code revision and default hyperparameters: this matters for the interpretation of R against the other runs (slide 13).

End-to-End Pipeline



Inference: fold pipelines and checkpoints are loaded, predictions averaged over folds and seeds, blended with the nested-CV weights; control rows set to 0.

Models and Preprocessing

TabularResNet

- Residual MLP with batch norm and dropout.
- RankGauss → PCA (50 gene and 15 viability components) concatenated to the features.

FT-Transformer [2]

- One token per feature, [CLS] token, self-attention.
- Simplified variant (post-LayerNorm, GELU).
- RankGauss only, no PCA.

CatBoost [3]

- `MultiLogloss`, 250 iterations, one model per fold.
- RankGauss only, no PCA.
- Targets with no training positive: smoothed Jeffreys prior.

Shared training choices

- `BCEWithLogitsLoss` with label smoothing 0.001 [8]; AdamW [6]; `ReduceLROnPlateau`; mixed precision [7].
- Early stopping (patience 8, at most 35 epochs) on an inner 10% split built with the *same grouping rule* as the outer folds; the outer fold is scored once.
- Fold-wise transforms (variance filter, RankGauss, PCA) are fitted on the training fold only [12].

Cross-Validation Protocols

All protocols: 5-fold iterative multilabel stratification [4, 5] on treated rows; the compound id comes from `train_drug.csv`.

Protocol	Unit kept together	Main property	Nested blend
`row`	single row	99.4% of validation rows have their compound in the training fold.	0.01595
`grouped`	whole compound	No overlap, but compounds have up to 718 rows: fold sizes deviate up to 25.6%.	0.01926
`hybrid`	compound, except those with $>$ 18 rows (split by row)	9 compounds (9.8% of rows) split by row; fold sizes within 1.4%; 9.8% of validation rows share their compound with training.	0.01715

blend: same code and tuned hyperparameters (runs R3, G3, H2).

Nested

Why hybrid?

Pure grouping is unbalanced and pessimistic for targets dominated by a few very large compounds; hybrid trades compound separation for balanced folds. The threshold is a *design choice*, not a proven optimum.

Choosing the Hybrid Threshold

Simulation on the real labels (one splitter seed): compounds with *more* rows than the threshold are split by row.

Scheme	Rows split by row	Max fold-size dev.	Val. rows with seen compound
pure grouped	0%	25.6%	0%
hybrid, $>$ 6 rows	22.0%	3.4%	22.0%
hybrid, $>$ 12 rows	11.9%	3.1%	11.9%
hybrid, $>$ 18 rows	**9.8%**	**1.4%**	**9.8%**
hybrid, $>$ 30, 60 or 100 rows	9.7%	2.8%	9.7%

- The rule: the *largest* threshold that brings fold sizes within a few percent keeps as much of the compound-aware protocol as possible.
- Thresholds 30, 60 and 100 give identical splits (8 compounds); the difference between 1.4% and 2.8% is within what another seed could change.
- The sensitivity of the CV *loss* to the threshold was not measured.

Ensemble and Nested Blending

- Predictions averaged over 3 seeds (and folds) for each network; CatBoost: one model per fold.
- Weights on the simplex via softmax reparameterisation, optimised with Nelder–Mead [11] on out-of-fold predictions.
- **Nested estimate**: weights fitted on four folds, score measured on the fifth, rotating the held-out fold.

$$\hat{p}_{\text{ens}} = w_R \hat{p}_R + w_F \hat{p}_F + w_C \hat{p}_C, \qquad w_R + w_F + w_C = 1$$

Final weights (ResNet / FT / CatBoost)

R: 0.453 / 0.375 / 0.173
R3: 0.679 / 0.264 / 0.057
H2: 0.478 / 0.348 / 0.175
G3: 0.489 / 0.151 / 0.360

Optimism of the blending step

In-sample minus nested score: $1.6 \cdot 10^{-6}$ (R), $2.6 \cdot 10^{-6}$ (H2), $4.5 \cdot 10^{-5}$ (G3, unequal folds): small against the difference between protocols.

The nested estimate limits optimism from fitting the weights; it does not remove bias from the CV design or from model selection.

Hyperparameter Tuning

Optuna [9] with TPE sampler [10] and median pruner, on compound-aware (hybrid) folds with their own split seed; the default configuration is trial 0 of every study.

Stage	Fold	What is done	Model selection in a trial
1, search	0	25/15/15 trials (ResNet/CatBoost/FT), one seed per trial	10% compound-aware inner split; the outer fold is scored once
2, confirmation	1	best 2 candidates and the default, 3 paired seeds	as above

- **Adoption rule**: gain $\geq 10^{-4}$ on the confirmation fold *and* wins on more than half of the 3 paired seeds. The search-fold score is the minimum over N trials, hence biased, and is not used.
- No FT-Transformer candidate beat the default: it keeps default values.
- An earlier exploratory cycle on row-wise folds (notebook not preserved, no result reported) informed the search space: it counts among the sources of optimism of the tuned runs.

Selected Hyperparameters

Model	Parameter	Default	Tuned (H2, R3, G3)
ResNet	Learning rate	10^{-3}	$2.67 \cdot 10^{-3}$
	Dropout	0.2	0.583
	Weight decay	10^{-5}	$2.66 \cdot 10^{-4}$
	Hidden units / batch	512 / 256	768 / 512
FT-Transformer	all parameters	default	Default
CatBoost	Depth / L2	5 / 3	4 / 56.0
	LR / subsample	0.08 / –	0.1034 / 0.829

- ResNet dropout (0.583), batch size (512) and learning rate ($2.67 \cdot 10^{-3}$) lie at or near the search-space bounds, and so does the CatBoost L2.
  The optimum may be outside the search space.
- The values were selected on hybrid folds: they may favour the hybrid protocol over row-wise and grouped.

Results: Cross-Validation

Out-of-fold log loss on treated rows (lower is better); networks use seed-averaged predictions.

Model	R: row/defaults	R3: row/tuned	H2: hybrid/tuned	G3: grouped/tuned
Baseline (per-target prior)	0.02220	0.02220	0.02233	0.02370
TabularResNet	0.01632	0.01606	0.01735	0.01971
FT-Transformer	0.01643	0.01642	0.01744	0.01992
CatBoost	0.01699	0.01701	0.01783	0.01979
Nested ensemble	**0.01607 ± 0.00017**	**0.01595 ± 0.00016**	**0.01715 ± 0.00031**	**0.01926 ± 0.00099**

- Ensemble improves on the best single model by about 1.5% (R), 0.7% (R3), 1.2% (H2) and 2.3% (G3).
- **Each model contributes under hybrid** (H2, drop-one nested blend): without ResNet $+1.2 \cdot 10^{-4}$, without FT $+0.8 \cdot 10^{-4}$, without CatBoost $+0.6 \cdot 10^{-4}$; removing any one worsens the score in all five folds.
- **Under grouped (G3) the FT-Transformer adds nothing**: $+1 \cdot 10^{-5}$ when removed (weight 0.15), against $+2.2 \cdot 10^{-4}$ (ResNet) and $+2.1 \cdot 10^{-4}$ (CatBoost); no model helps in all five folds.
- Residual correlations are high (0.98 between networks); CatBoost is the most different on positive cells (0.945 in H2, 0.81 in G3).

Row-Wise versus Compound-Aware CV

Controlled comparison: R3, H2, G3

Same code, seeds and tuned hyperparameters; only the folds change. R (defaults, earlier code) is not part of it: R to H2 (+6.7%) and R to G3 (+19.9%) also reflect hyperparameters and code.

	R3 (row)	H2 (hybrid)	G3 (grouped)
Validation rows with seen compound	99.4%	9.8%	0%
Nested blend (against H2)	0.01595 (−7.0%)	0.01715	0.01926 (+12.3%)

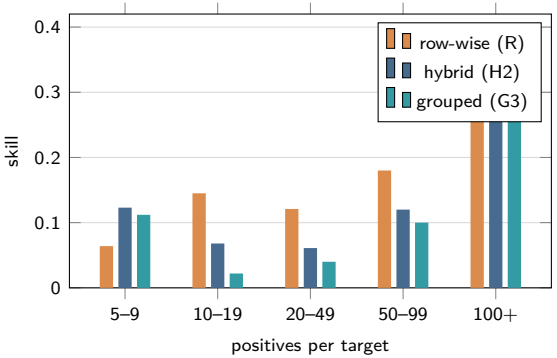
Paired decomposition by compound frequency (blend, bootstrap over compounds; diagnostics, section 7):

Rows	Share	R3	H2	G3
Frequent compounds	9.8%	0.00165	0.00198	0.02260
All other rows	90.2%	0.01750	0.01879	0.01884

- Ordinary compounds carry +0.00116 of the +0.00119 from R3 to H2 (+7.4%, 95% CI [+0.00116, +0.00143]); the frequent ones carry +0.00201 of the +0.00206 from H2 to G3.
- Under grouped, targets carried by few compounds lose most skill against R: `hmgcr_inhibitor` (283 positives, 7 compounds) 0.834 → 0.217.

Per-Target Analysis (blend)

Skill $= 1 -$ loss/loss of the per-target prior, by number of positives (row-wise, hybrid, grouped):



The per-target baseline differs between protocols: skills are not strictly comparable. Bin 1–4 (2 targets) omitted.

- The 41 targets with ≥ 100 positives carry 47.9% of the loss (skill 0.342).
- The 53 *rare* targets (< 20 positives) carry 6.6% of the loss; skill $+0.080$ [0.056, 0.102] in H2 (G3: $+0.044$ [-0.013, 0.085]); for 8 of them the model does not beat the prior.
- The blend over-predicts rare targets: expected/observed positives 1.47 below 20 positives.
- 31 targets with ≥ 10 positives come from at most 3 compounds: close to compound-identity labels.

Independent Check: Kaggle Test Set

Two late submissions through the inference notebook (runs R and H2); the CV estimates of R3 and G3 are compared with the same scores.

	R	R3	H2	G3
Nested-CV blend, treated rows	0.01607	0.01595	0.01715	0.01926
Rescaled to all test rows ($\times 0.910$)	0.01462	0.01452	0.01561	0.01753
Kaggle private	0.01634	–	0.01643	–
Kaggle public	0.01865	–	0.01873	–
Private (R / H2) above rescaled CV	+11.8% / +12.4%	+12.5% / +13.2%	+4.7% / +5.3%	−6.8% / −6.3%

- **Both private scores lie between the hybrid and the grouped estimates**: hybrid CV is slightly optimistic, grouped CV pessimistic (it reproduces 169% of the gap from the row-wise estimate, hybrid 57%). Which is closer depends on the unknown overlap of ordinary compounds: indirect evidence only.
- The two submissions differ by $9 \cdot 10^{-5}$ (private) and $8 \cdot 10^{-5}$ (public): **they cannot be ranked**; protocol, tuning and model selection did not move the test score measurably.
- Public and private scores differ by 0.0023 for *both*: the test subsets differ in difficulty.

The test set was never used to tune or select; it did motivate the compound-aware protocols. The rescaling uses the public treated share (0.910): approximate. R3 and G3 were never submitted.

Do the Test Compounds Overlap with Training?

Indirect analysis from nearest-neighbour cosine distances (no compound id in the test set).

- **General overlap: not assessable.** Distance separates seen from unseen compounds poorly (AUC 0.516); the mixture estimate must be discarded.
- **Frequent compounds** (5 of 8 analysed are present):

Compound	Test rows assigned	Expected if absent	Test share	Train share
87d714366	133	5.3	3.93%	3.27%
8b87a7a83	23	0.5	0.69%	0.92%
d1b47f29d	24	0.5	0.72%	0.81%
d08af5d4b	36	11.4	0.83%	0.89%
292ab2c28	18	6.2	0.38%	0.88%

- Four are at about their training frequency; 292ab2c28 at about half of it.
- For the other three the assigned rows are close to those expected if absent: presence cannot be established.
- A back-of-the-envelope reweighting with the hybrid losses gives an all-rows estimate of 0.0156–0.0161 against 0.01643 private (residual gap 2–5%): inference, not measurement.

Negative Results

Nearest-neighbour override for frequent compounds

- Pull predictions of rows assigned to a frequent compound towards its label profile (parameters from the training part of each fold).
- Rule precision 0.722, recall 0.809; gain ceiling only $1.7 \cdot 10^{-4}$.
- All four variants **worsen** the OOF loss on all five folds: from $+2.1 \cdot 10^{-4}$ (mixture 0.5) to $+1.47 \cdot 10^{-3}$ (replacement).
- Cost sits on false assignments (663 of 2,381 rows).

Extra networks (run R protocol)

Configuration	Ensemble	Extra weight
+ MLP (PCA)	0.01604	0.015
+ 1D-CNN (PCA)	0.01605	0.082
+ 1D-CNN (raw)	0.01606	0.070
no extra network	0.01605–0.01606	–

Differences $\leq 2 \cdot 10^{-5}$: below the fold standard deviation (0.00017) and comparable to run-to-run noise ($\approx 10^{-5}$).

Takeaway

More components or compound-specific post-processing gave no robust gain on the available validation evidence: the simpler three-model ensemble was kept.

Limitations

1. **Test compound ids unknown:** overlap is inferred only for frequent compounds; ordinary-compound overlap is unassessed, so neither hybrid nor grouped CV can be declared the faithful one (they bracket the private score).
2. **Hybrid threshold is heuristic:** 18 rows chosen from fold balance for one seed; CV-loss sensitivity not measured.
3. **Comparison of protocols:** only R3, H2 and G3 are comparable (one run each); R used defaults and an earlier code revision. The tuned values were chosen on hybrid folds and may favour that protocol.
4. **Grouped CV is unbalanced:** folds of 3933–5512 rows, 34–36 targets without validation positives, noisier nested score; a pessimistic bound, not a faithful estimate.
5. **Selection bias:** tuning and model choices reused folds overlapping the reported OOF; no untouched hold-out.
6. **Limited tuning:** one seed per trial, few trials, optimum possibly outside the ResNet search space; an earlier exploratory cycle (row-wise folds, not preserved, not reported) informed the search space; no controlled tuned-versus-default comparison.
7. **Preprocessing:** unsupervised transforms include inner-validation rows; may slightly affect early stopping.
8. **Test evidence:** two submissions, noisy and different public/private subsets; fold standard deviation (0.0003) understates the uncertainty on unseen data.
9. **Reproducibility:** GPU non-determinism ($\approx 10^{-5}$); inference needs the separately saved artifact dataset.

Conclusions

- A reproducible training/inference workflow for a sparse 206-target multi-label task, with separated notebooks and saved artifacts.
- Compound-aware CV exposes a harder generalisation setting than row-wise CV: with the same code and hyperparameters the nested loss goes from 0.01595 (row) to 0.01715 (hybrid) and 0.01926 (grouped), and the two compound-aware protocols bracket the private score.
- The three-model ensemble beats each component in the nested-CV experiments; under hybrid CV each model helps in all five folds, under grouped CV the FT-Transformer adds nothing.
- Tuning did not measurably improve the hybrid ensemble ($-5 \cdot 10^{-5}$ against defaults) and helped the grouped one slightly ($-3.1 \cdot 10^{-4}$, not proven); the two Kaggle submissions cannot be ranked against each other.
- Nearest-neighbour post-processing and extra networks brought no gain: simplicity was preferred.

Main lesson

For repeated-compound biological measurements, validation design can matter at least as much as model architecture. A CV score must be read together with the assumptions of its split, here the unknown overlap between test and training compounds.

References


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